Final Exam – Medicinal Plants (Very Hard Level)

Essay-Based Examination

Syllabus: Lectures 5–12

Time: 3 Hours

Section A – Very Hard Essay Questions

1. A patient with chronic constipation and hypokalemia is using herbal medicines regularly. Discuss the risks associated with Senna use in this patient, explaining the mechanism of action, toxicity, and safer alternatives.

2. Explain why Uva Ursi is unsuitable for long-term therapy. Include biochemical activation, site of action, and potential toxicity.

3. Discuss the toxicological profile of Boldo leaves, explaining the role of volatile oil constituents and clinical consequences.

4. Artichoke is used in hyperlipidemia management. Discuss its mechanism of action and explain how this relates to bile secretion and cholesterol metabolism.

5. Compare Senna, Guava, and Ivy regarding their pharmacological actions, therapeutic uses, and risks of misuse.

6. A child presents with productive cough. Critically evaluate the suitability of Ivy leaves, including mechanism, contraindications, and precautions.

7. Explain the pharmacological and toxicological effects of Eucalyptus oil following oral ingestion.

8. Discuss Tea leaves as CNS stimulants, focusing on active constituents, mechanism, and adverse effects.

9. Hyoscyamus toxicity presents as a classical syndrome. Describe this syndrome, its manifestations, and underlying mechanism.

10. Discuss the therapeutic importance of Jaborandi leaves, explaining the pharmacological action of pilocarpine.

11. Explain the dual pharmacological nature of Coca leaves and discuss risks associated with misuse.

12. Digitalis is a life-saving but dangerous drug. Discuss its mechanism of action, narrow therapeutic index, and factors increasing toxicity.

13. Explain the relationship between hypokalemia and increased Digitalis toxicity at the molecular level.

14. Discuss Ginkgo leaves in the management of cerebral circulation disorders, including drug–herb interactions.

15. Explain how plant taxonomy contributes to safe medicinal plant use, giving examples of toxic plants.

16. Compare gymnosperms and angiosperms with respect to seed structure and medicinal relevance.

17. Discuss the importance of proper plant identification in preventing adulteration and toxicity.

18. Explain the role of saponins in respiratory therapy, with reference to Ivy leaves.

19. Discuss chronic laxative abuse from medicinal plants, including physiological consequences.

20. Explain the difference between hepatoprotective and choleretic actions, giving suitable medicinal plant examples.

Section B – Model Answers

1. Senna contains anthraquinone glycosides that stimulate colonic peristalsis and inhibit water reabsorption. In hypokalemia, Senna increases potassium loss, predisposing to arrhythmias. Safer alternatives include bulk laxatives.

2. Uva Ursi contains arbutin, which is converted to hydroquinone in alkaline urine. Long-term use leads to hydroquinone toxicity and renal irritation.

3. Boldo contains ascaridole, a toxic monoterpene peroxide causing hepatotoxicity and neurotoxicity.

4. Artichoke contains cynarin which increases bile secretion, enhancing cholesterol elimination.

5. Senna is a stimulant laxative, Guava is astringent due to tannins, and Ivy is an expectorant due to saponins.

6. Ivy contains triterpenoid saponins that liquefy mucus; however, it is contraindicated in infants.

7. Oral Eucalyptus oil causes GI irritation and CNS depression leading to respiratory failure.

8. Tea contains xanthines such as caffeine causing CNS stimulation and insomnia.

9. Hyoscyamus causes anticholinergic syndrome characterized by dry mouth, mydriasis, and hallucinations.

10. Jaborandi contains pilocarpine, a muscarinic agonist used in glaucoma.

11. Coca leaves act as CNS stimulants but can lead to addiction and cardiovascular toxicity.

12. Digitalis increases intracellular calcium but has a narrow therapeutic index.

13. Hypokalemia increases binding of Digitalis glycosides to Na^+/K^+-ATPase.

14. Ginkgo improves cerebral blood flow but increases bleeding risk with anticoagulants.

15. Taxonomy ensures correct identification preventing toxic substitution.

16. Gymnosperms have naked seeds, angiosperms have covered seeds.

17. Proper identification prevents adulteration and poisoning.

18. Saponins reduce mucus surface tension aiding expectoration.

19. Chronic laxative abuse causes dependency and electrolyte imbalance.

20. Hepatoprotective agents protect liver cells, choleretic agents increase bile flow.